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Analysis

The Impact of Antibiotic-Mediated Modification of the Intestinal Microbiome on Outcomes of Allogeneic Hematopoietic Cell Transplantation: Systematic Review and Meta-Analysis



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A B S T R A C T

Accumulating evidence points toward a protective role of intestinal microbiota diversity in allogeneic hematopoietic cell transplantation (allo-HCT). The purpose of this systematic review and meta-analysis is to determine the effect of antibiotic-mediated disruption of microbiota on main allo-HCT outcomes (graft-versus-host disease [GVHD], treatment-related mortality [TRM], overall survival [OS]). Following literature search, 2 reviewers screened eligible studies and assessed risk of bias (RoB). Meta-analysis was performed using Review Manager Software. Among 443 screened references, 18 were eligible for meta-analysis. In studies with genomic markers, grade II to IV acute GVHD was significantly reduced in patients not receiving gut decontamination (GD) (odds ratio [OR], 1.56; 95% confidence interval [CI], 1.20 to 2.04). In subgroup analysis, prophylaxis with systemic antibiotics conferred an increased risk of acute GVHD (OR, 1.65; 95% CI, 1.08 to 2.53). When we incorporated RoB, we found a positive correlation of intestinal GVHD with GD (OR, 1.77; 95% CI, 1.29 to 2.44). Patients with higher microbiota diversity presented increased OS (risk ratio [RR], 1.58; 95% CI, 1.19 to 2.09) and lower TRM (RR, 0.45; 95% CI, 0.26 to 0.76). Our findings confirm that GD and prophylaxis with systemic antibiotics increase acute and intestinal GVHD. Importantly, our meta-analysis was the first to show a significant effect of microbiota diversity on TRM and OS.

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Allogeneic hematopoietic stem cell transplantation (allo-HCT) is a potentially curative treatment for a variety of diseases, ranging from congenital disorders to malignancies [1]. Despite significant improvements in conditioning regimens and supportive care, graft-versus-host-disease (GVHD) remains a major cause of morbidity and mortality, associated not only with the disease per se but also with other complications [2–5]. Therefore, efforts of better understanding and managing GVHD represent a continuous unmet clinical need in the field of allo-HCT [6].

In the 1970s, van Bekkum et al. [7], and Jones et al. [8] performed pioneering work in the pathophysiology of GVHD, demonstrating that it was rarely evident in animals free of microbes. These experimental studies led to clinical application of gut decontamination (GD) strategies aiming to achieve a protective environment [9,10]. Despite initial promising results, further studies were not able to confirm this benefit [11–13]. Beelen et al. [14] showed that intestinal anaerobic

bacteria are involved in the pathogenesis of acute GVHD. However, further experimental studies of prophylactic administration of probiotic lactobacilli changed this paradigm [15]. This led to revolutionary studies applying novel genomic tools (16S ribosomal RNA [16S rRNA] and metagenomics), allowing in-depth understanding of bacterial diversity [16].

In the era of precision medicine [17], identifying patients at risk for GVHD and adverse outcomes is a major goal. Novel genomic evidence points toward a predictive role of intestinal microbiota as a biomarker of GVHD [18]. Nevertheless, there is a difficulty of method standardization and heterogeneity in microbiota diversity scoring. In addition, there is no systematic validation of effective antibiotic approaches that preserve microbiota diversity [19,20]. Importantly, approaches to re-establish microbiota diversity remain under investigation [21].

Given the lack of systematic reviews and/or meta-analyses in the field, the purpose of this systematic review and meta-analysis is to determine the effect of the intestinal microbiome modification caused by antibiotics on main allo-HCT outcomes (GVHD, treatment-related mortality [TRM], overall survival [OS]). Hence, the primary objective is to assess the impact of antibiotics on GVHD, TRM, and OS. The secondary objective is to clarify the association of antibiotic-mediated disruption of microbiota as indicated by microbiome markers (diversity

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indexes such as 3-indoxyl sulfate [3-IS], Simpson, inverse Simpson, Shannon, Chao1, or stool cultures) with outcomes in allo-HCT recipients.

METHODS

This systematic review and meta-analysis is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines (PRISMA; Supplementary Table S1) [22]. We predetermined the study protocol and registered it online at PROSPERO database (ID: CRD42019126184). We conducted an extensive systematic literature search through March 25, 2020. Our search strategy is available online, along with an extensive report of our methods in the Supplementary Methods [23].

Risk of Bias Assessment

Two reviewers (M.G., E.G.) independently assessed the quality of included studies. We assessed risk of bias (RoB) in cohort studies using the Newcastle-Ottawa scale [24] and in randomized trials using the Cochrane risk of bias assessment tool [25]. We assessed publication bias visually by funnel plots. We also evaluated the certainty in our estimates applying the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) tool for our primary outcome [26].

Summary Measures and Synthesis of Results

The primary outcome of our meta-analysis was estimated by comparing groups of different antibiotics according to comparisons of original studies. Since most studies compared groups that achieved GD to selective or no decontamination, we followed this terminology. Thus, we estimated odds ratios (ORs) of these populations to develop acute or chronic GVHD; we planned to assess also the risk ratio (RR) of OS and TRM whenever those data were available. All outcomes are presented with 95% confidence intervals (CIs).

Our secondary endpoint aimed to assess the association between antibiotic-mediated fecal microbiota suppression and allo-HCT outcomes (GVHD, OS, TRM). Recent advances in the field allowed the study of patients' fecal community through bacterial 16S rRNA, clustering into operational taxonomic units or metagenomic sequencing [27,28]. We extracted relevant information about bacterial fecal species in our systematic review of literature. More important, we tried to quantify this association using diversity indices (ie, Shannon index, Simpson index, and inverse Simpson index) reported to significantly affect outcomes of interest (GVHD/OS/TRM) [29,30].

Sensitivity and Subgroup Analyses

We preplanned to perform a sensitivity analysis integrating RoB assessment. We also conducted subgroup analyses according to (1) age of population (pediatric or adult), (2) microbiota marker (nongenomic based on cultures and genomic based on genetic studies), (3) GVHD type or grade, and (4) systemic versus topical antibiotics.

RESULTS

Study Selection—Flow Diagram

A total of 443 references were imported for screening (301 were identified through the first literature search, 84 through the second, 58 through the third, and 17 additional records identified through other sources), as illustrated by a PRISMA flow diagram in Figure 1 [31]. We deemed 26 studies eligible for qualitative synthesis, 18 of which were included in the meta-analysis.

Descriptive Characteristics

Characteristics and RoB assessment of each study included in this systematic review are presented in Table 1 and Supplementary Tables S2 to S3.

The impact of GD on GVHD or OS has been analyzed in 15 studies in the literature. In 1989, the first randomized controlled trial (RCT) in the field compared the efficacy of 2 quinolones (norfloxacin versus pefloxacin) in decontaminating intestinal microbiota [32]. Pefloxacin administration markedly reduced *Bacteroides* spp., while no patient developed GVHD. In 1999, the second RCT proved that a combination of metronidazole and ciprofloxacin as prophylaxis resulted in a significant reduction of anaerobic bacteria fecal culture growth, reduced acute and chronic GVHD, and extended 5-year OS [14]. The advantage of complete over selective GD on GVHD occurrence was supported by Vossen et al. in a pediatric

retrospective cohort in 1990 [10] and 2014 [33], using surveillance stool cultures. In the earlier study, Vossen et al. concluded that the risk of acute GVHD was significantly reduced in the complete GD group. In the recent study, patients who did not achieve complete GD had improved both relapse-free and overall survival, but these results were not statistically significant.

Further clinical studies changed our perception of GD. First and foremost, Taur et al. [34] in 2014 recognized fluoroquinolone administration and low diversity of fecal flora (defined as inverse Simpson lower than 2) as independent predictors of decreased OS in a retrospective analysis of stool samples collected at neutrophil engraftment. Shono et al. [35] studied the effect of various antibiotics for febrile neutropenia and documented that broad-spectrum antibiotics not only led to perturbation of gut microbiota composition but also increased GVHD incidence and TRM. Weber et al. [36] published several studies in the field with valuable conclusions. Primarily, they revealed the prognostic impact of antibiotic-related low urinary 3-IS levels as a microbiota marker of higher TRM and worse OS. In a recent retrospective adult cohort study, they proved that prophylactic use of rifaximin instead of ciprofloxacin plus metronidazole correlated with lower enterococcal positivity and higher urinary 3-IS, showing improved allo-HCT outcomes [37]. They also highlighted timing of antibiotic treatment as the dominant independent risk factor for TRM and reduction of Clostridiales [38]. In 2018, Weber et al. [39] confirmed the abovementioned superiority of rifaximin in urinary 3-IS levels, Shannon index, and allo-HCT outcomes, after adjustment for confounding factors, such as treatment with systemic broad-spectrum antibiotics. Taking advantage of breakthrough techniques, Kaysen et al. [40] used both metagenomic and meta-transcriptomic sequencing in a prospective cohort study. In addition to a detrimental effect on bacterial diversity, antibiotic treatment resulted in higher expression of antibiotic resistance genes, adversely affecting GVHD and survival. In the light of these discoveries, Han et al. [41] confirmed in 2019 that microbiota diversity assessed by inverse Simpson index was lower in the acute GVHD group and negatively associated with β -lactam antibiotic administration. The authors took a step forward in a similar study, which in our analysis is combined with the abovementioned one, proposing a score to universally predict severe acute GVHD with an area under the curve of 0.81 [42]. Last, Lee et al. [43] investigated the impact of the spectrum of antibiotics during the neutropenic period. Thus, broad-spectrum antibiotics were associated with loss of microbiota diversity and increased intestinal GVHD but not with higher TRM or worse OS. Interestingly, intestinal GVHD was also associated with decreased abundance of *Ruminococcus* and increased abundance of *Eubacterium* and *Escherichia* [43].

The above-described findings also apply to pediatric allo-HCT recipients. Simms-Waldrup et al. [44] conducted a prospective analysis verifying a significant decline in gut *Clostridia* of patients with GVHD. GVHD was strongly associated with the use of antibiotics, particularly those effective against anaerobic bacteria. In a small cohort followed longitudinally, Bekker et al. [45] found no difference in acute GVHD between patients receiving selective or total GD. The authors, however, observed that during selective GD, microbiota composition was dominated by *Bacteroides*, in contrast with genera *Enterococcus* and *Streptococcus* blooming at total GD group. Furthermore, an ongoing randomized phase 2 study, which was published as an interim analysis in a conference abstract, examines the effect of oral vancomycin-polymyxin as GD therapy on microbiota diversity and acute GVHD. Microbiota composition was measured by Shannon index using shotgun

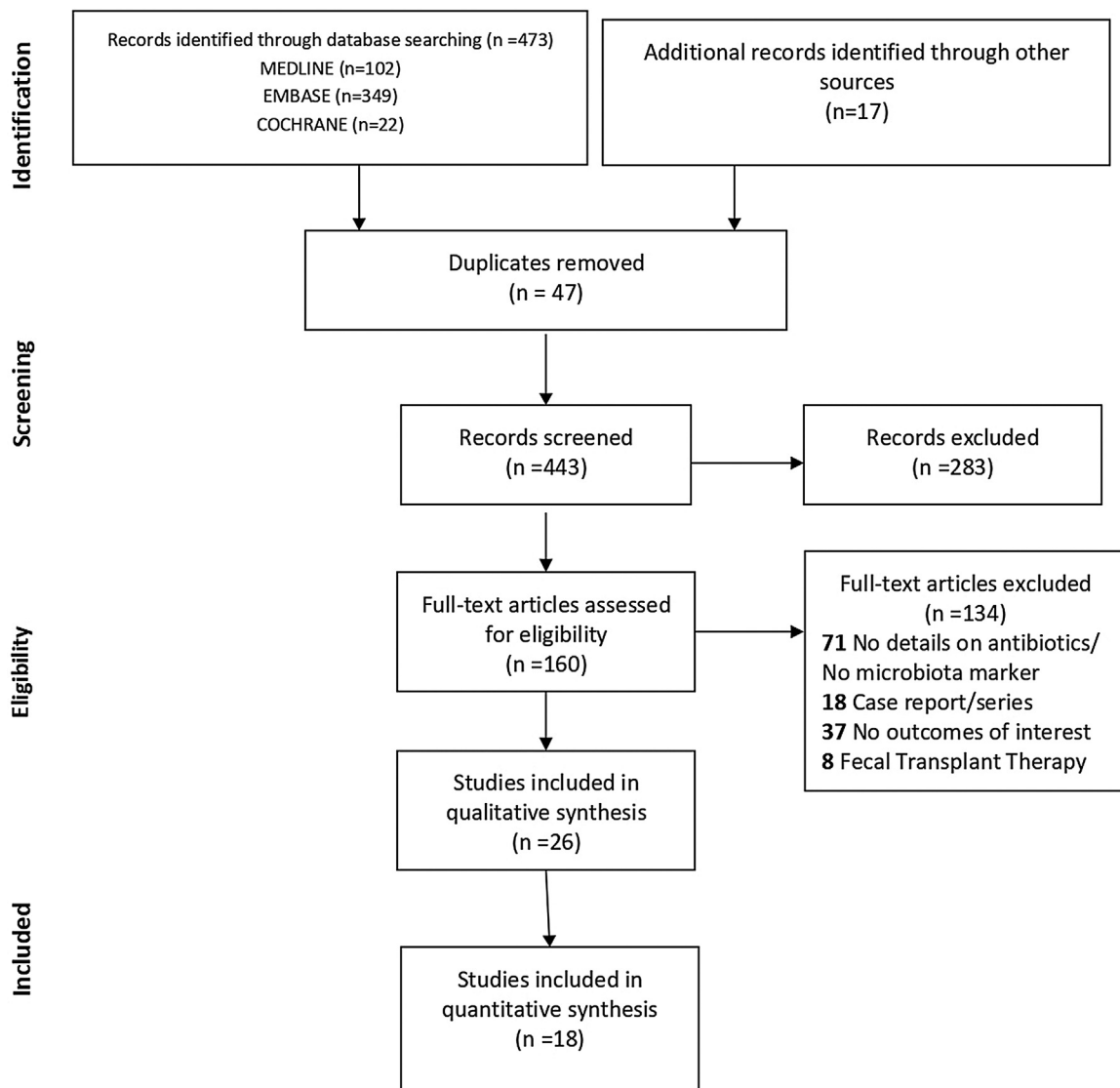


Figure 1. The PRISMA flow diagram summarizes search, selection process, and reasons for exclusions through different phases of systematic review.

metagenomic sequencing. GD group was associated with higher average change in Shannon index at 2 weeks [46].

The remaining studies included in this systematic review address issues of antibiotic-induced disruption of microbiota, the impact on bacteria species, and their association with allo-HCT outcomes. In the largest study, Peled and Gomes [47] used the inverse Simpson index during the periengraftment period to classify patients into high- or low-diversity groups. Lower diversity was related to a higher risk of TRM and reduced OS (even when exposure to piperacillin-tazobactam and meropenem was incorporated to multivariable models). Domination of a single taxon, *Enterococcus*, was observed across all centers. Peled and Gomes [47] produced a risk score to identify a microbiota signature predictive of survival. In a prospective cohort study, Jenq et al. [48] found that patients with GVHD had increased abundance of Lactobacillales, reduction of Clostridiales, and greater bacterial chaos early after allo-HCT. In another clinical study, Jenq et al. [49] underlined the role of a specific clostridial species—*Blautia* abundance in improved OS, reduced acute GVHD grade II to IV, and GVHD-related mortality (area under the curve = 0.72). In a similar study, Biagi et al. [50] confirmed that reduced diversity

expressed by a lower Simpson index, lower *Blautia*, and increased *Fusobacterium* abundance at the time of engraftment were also predisposing factors of acute GVHD in children. Recently, Kusakabe et al. [51,52] published 2 studies in the field; a combination of their data is presented in tables and used in our meta-analysis. At first, Kusakabe et al. [51] used metagenomic analyses to examine time-dependent alterations of microbiota's beta diversity in allo-HCT recipients, compared to healthy controls and autologous HCT recipients. They associated loss of diversity with 2-year OS. Interestingly, when methods for α -diversity evaluation (Simpson index) were used, no difference was identified among 3 groups [51]. In an effort to examine the role of each bacterial genus, Kusakabe et al. [52] showed that *Enterococcus* abundance is related to poor OS and nonrelapse mortality; nevertheless, the result could not be replicated by conventional stool culture. Ingham et al. [53] studied a cohort of pediatric recipients and observed low abundances of Ruminococcaceae and high abundances of Lactobacillaceae taxa in patients with acute GVHD and high mortality. They proved that vancomycin and ciprofloxacin were associated with a shift at microbiome taxa. The disturbance of fecal environment (quantified by mean urinary 3-IS)

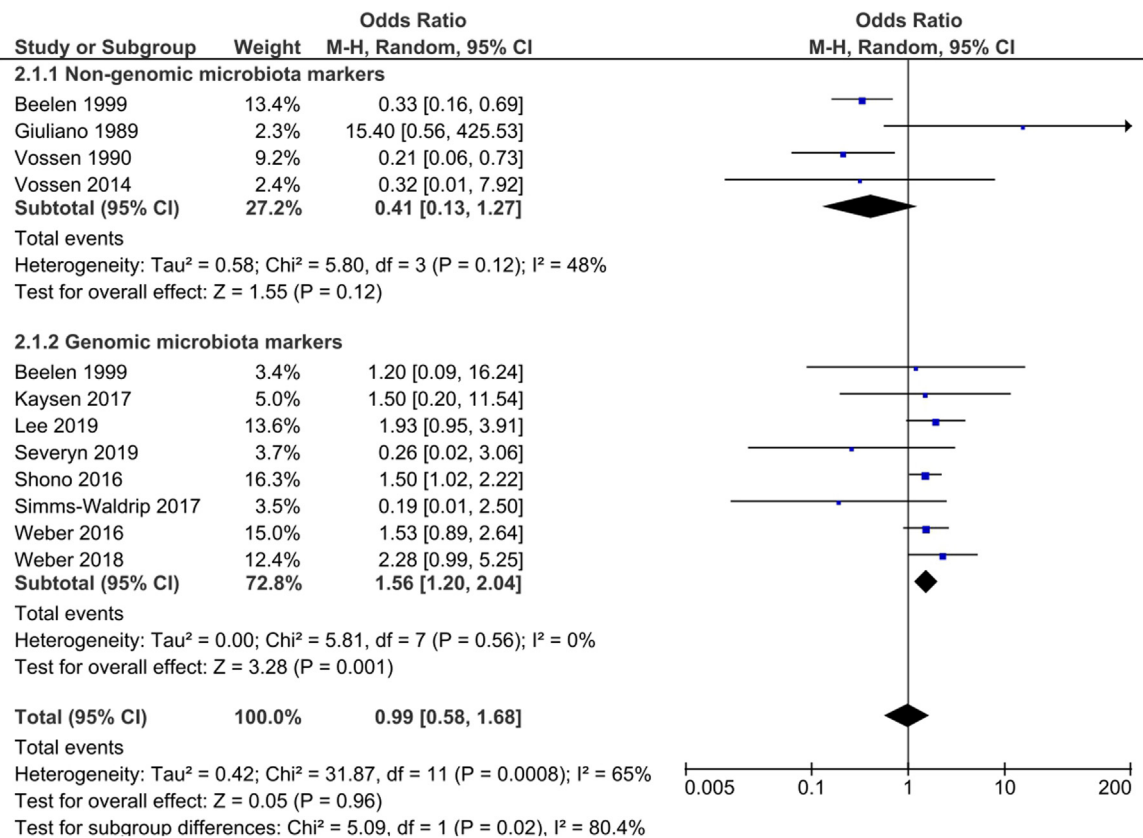


Figure 2. Forest plot of 12 studies with data on acute GVHD (aGVHD) occurrence in patients receiving or not (group versus control) antibiotics, taking into consideration microbiota diversity via genomic or nongenomic markers. Odds ratio and weight of each study are presented along with 95% confidence interval (CI). Analysis was made using the random-effects (RE) model. Diamonds represent summary of each comparison or subgroup analysis and indicate that even though there is no statistically significant difference among all studies, subgroup analysis of genomic markers showed that aGVHD occurred less frequently in the control group (not receiving antibiotics).

and increase in enterococci caused by broad-spectrum antibiotics resulted in increased intestinal GVHD [54]. A more recent prospective study assessed the value of the inverse Shannon index and balanced weighted phylogenetic diversity as microbiota markers predicting the incidence and severity of acute GVHD. The study proposed a gradient of bacterial species resulting from the difference of relative bacterial abundances as the most predictive variable of acute GVHD [18]. Liu et al. [55] prospectively compared microbiota composition of recipients with that of donors and identified high bacterial donor diversity as a protective factor for acute GVHD. Moreover, recipients with lower microbiota diversity had 63% higher overall mortality rates but no statistically significant difference in acute GVHD risk [55]. Furthermore, Galloway-Peña et al. [56] observed that the risk of intestinal acute GVHD was 45% lower with a higher Shannon diversity index at the time of engraftment, and Shannon index was a statistically significant risk factor of TRM. That was not the case for fecal metabolites (indole and butyrate levels).

Risk of bias within studies

The quality of included studies varied, as shown in Table 1 and Supplementary Tables S2 to S3.

GVHD Incidence

As presented in Figure 2, the odds of acute GVHD occurrence depended on the method used for microbiota diversity measurement. In 4 studies in which nongenomic markers were used (bacterial culture growth of feces), the GD group tended to have a

favorable outcome despite the lack of statistical significance (OR, 0.41; 95% CI, 0.13 to 1.27; $I^2 = 48\%$; $P = .12$). However, in 8 more recent studies that used genomic markers (such as chao1, inverse Simpson and Shannon diversity index, urinary 3-indoxyl sulfate levels, phylogenetic diversity using 16S/18S rRNA, chromatography/spectrometry or shotgun metagenomic sequencing), the occurrence of severe acute GVHD was significantly reduced in patients not receiving GD (OR, 1.56; 95% CI, 1.20 to 2.04; $I^2 = 0\%$; $P = .001$). Nevertheless, the overall estimate of our meta-analysis found no difference of grade II to IV acute GVHD incidence between the 2 groups due to the abovementioned difference (OR, 0.99; 95% CI, 0.58 to 1.68; $I^2 = 65\%$; $P = .96$). We performed a preplanned subgroup meta-analysis according to age of population, type of GVHD, and use of systemic antibiotics. Interestingly, when we incorporated RoB into the outcome of intestinal GVHD, we found a positive correlation of intestinal GVHD with GD (OR, 1.77; 95% CI, 1.29 to 2.44; $I^2 = 0\%$; $P = .004$), suggesting that patients not receiving antibiotics had a lower frequency of intestinal GVHD (Figure 3). However, pediatric studies showed an inverse association (Figure 4 and Supplementary Figure S1). GD in children was significantly associated with less severe acute GVHD (OR, 0.27; 95% CI, 0.11, 0.68; $I^2 = 0\%$; $P = .005$). We performed a sensitivity analysis to eliminate heterogeneity of the adults' subgroup analysis and proved that adults who did not receive GD had a lower risk of aGVHD (OR, 1.78; 95% CI, 1.21 to 2.61; $I^2 = 0\%$; $P = .003$), as shown in Figure 4.

Taking into consideration the descriptive characteristics of available studies, we found only 3 recent studies with low RoB that compared systemic versus topical or no antibiotics as

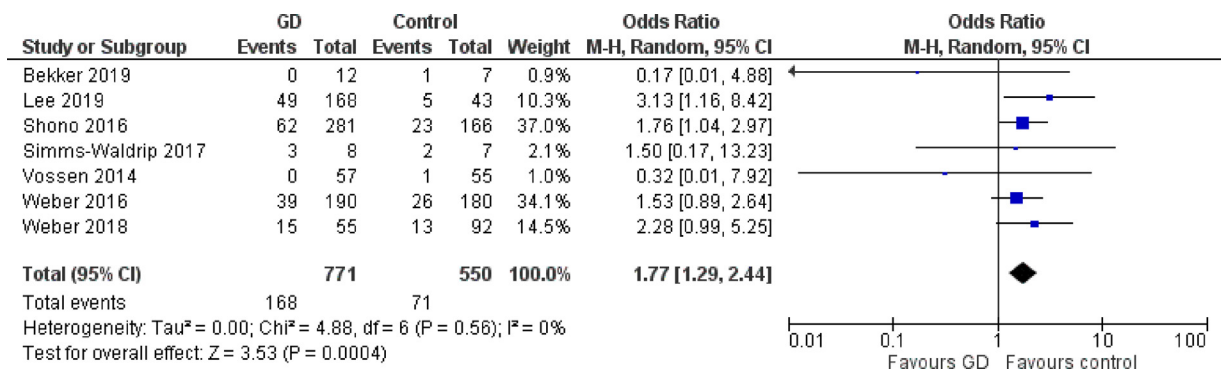


Figure 3. Forest plot of 7 studies presenting odds ratio of intestinal aGVHD occurrence in patients receiving or not (group versus control) antibiotics. Odds ratio and weight of each study are presented along with 95% CI. Analysis was made using the RE model. The diamond represents summary of results; intestinal aGVHD occurred less frequently in the control group (not receiving antibiotics).

prophylactic treatment [37,43,45]. As shown in Figure 5, patients receiving systemic antibiotics had a statistically significant greater risk of developing acute GVHD (OR, 1.65; 95% CI, 1.08 to 2.53; $I^2 = 0\%$; $P = .02$). It is important to point out that separately, each study failed to reach statistical significance and support the devastating effect of broad-spectrum antibiotics on acute GVHD.

Impact on survival

We aimed to clarify the impact of antibiotics on OS considering microbiota diversity. Patients who received antibiotics had a 10% reduction in OS (RR, 0.90; 95% CI, 0.77 to 1.06; $I^2 = 78\%$; $P = .23$) compared with controls, as presented in Figure 6. Nevertheless, heterogeneity remained high and the outcome was not

statistically significant, even after incorporation of RoB or subgroup analysis of genomic versus nongenomic microbiota methods. Similarly, we found that antibiotic administration increased the risk of TRM by 75% (RR, 1.75; 95% CI, 0.96 to 3.17; $I^2 = 75\%$; $P = .07$; Figure 7), which was not statistically significant.

Then, we performed a meta-analysis of 5 studies to evaluate the effect of microbiota diversity on OS, as shown in Figure 8. Patients with higher microbiota diversity presented increased OS compared with patients with lower diversity (RR, 1.58; 95% CI, 1.19 to 2.09; $I^2 = 79\%$; $P = .002$). Heterogeneity was moderate even after the incorporation of RoB ($I^2 = 51\%$; Supplementary Figure S2) due to the lack of a unanimous approach regarding the measurement of microbiota diversity.

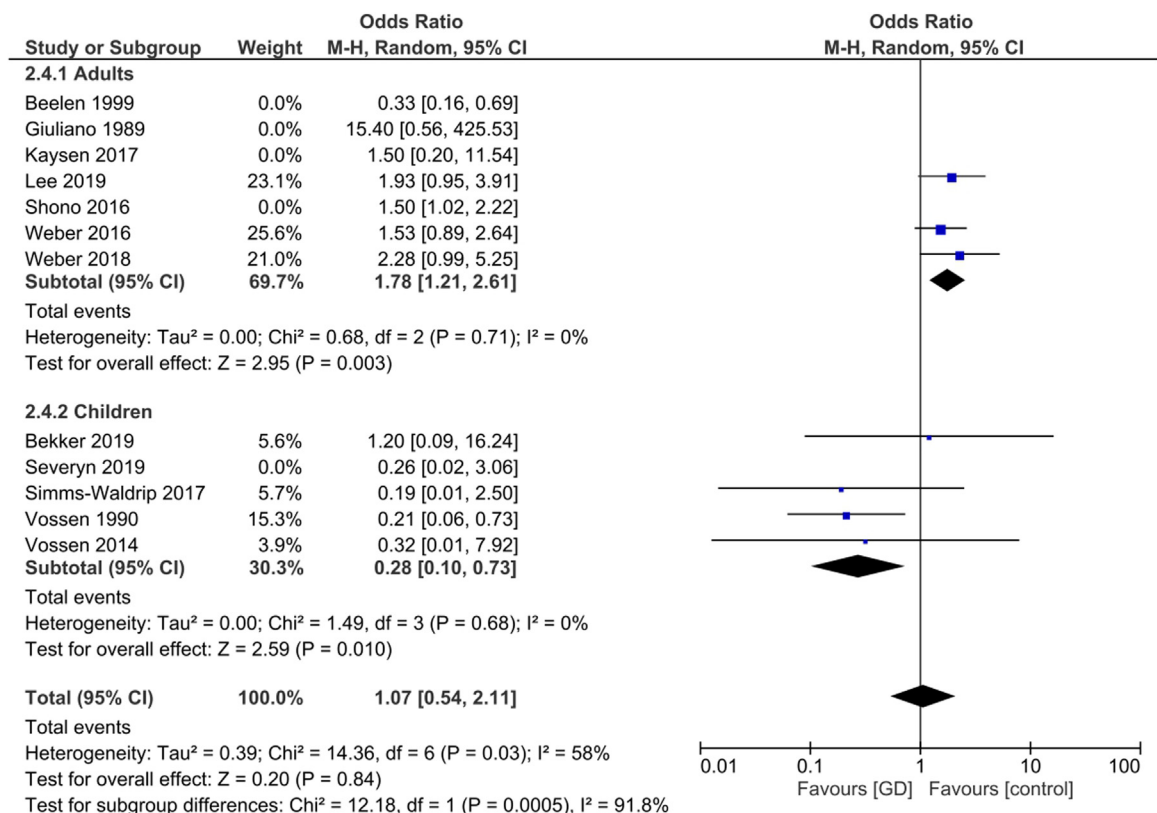


Figure 4. Forest plot of 7 low RoB studies with data on aGVHD occurrence in patients receiving or not (group versus control) antibiotics with subgroup analysis according to age. Odds ratio and weight of each study are presented along with 95% CI. Analysis was made using the RE model. Diamonds represent summary of each comparison or subgroup analysis and indicate that even though there is no statistically significant difference among all studies, subgroup analyses according to age showed that aGVHD occurred less frequently in children receiving antibiotics and more frequently in adults receiving antibiotics.

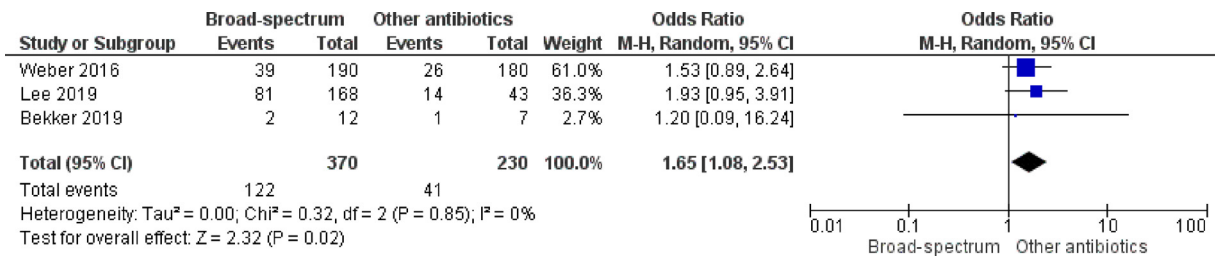


Figure 5. Forest plot of 3 low RoB studies with data on aGVHD in patients receiving systemic antibiotics versus others. Odds ratio and weight of each study are presented along with 95% CI. Analysis was made using the RE model. The diamond represents summary of results; aGVHD occurred less frequently in the control group (not receiving broad-spectrum antibiotics).

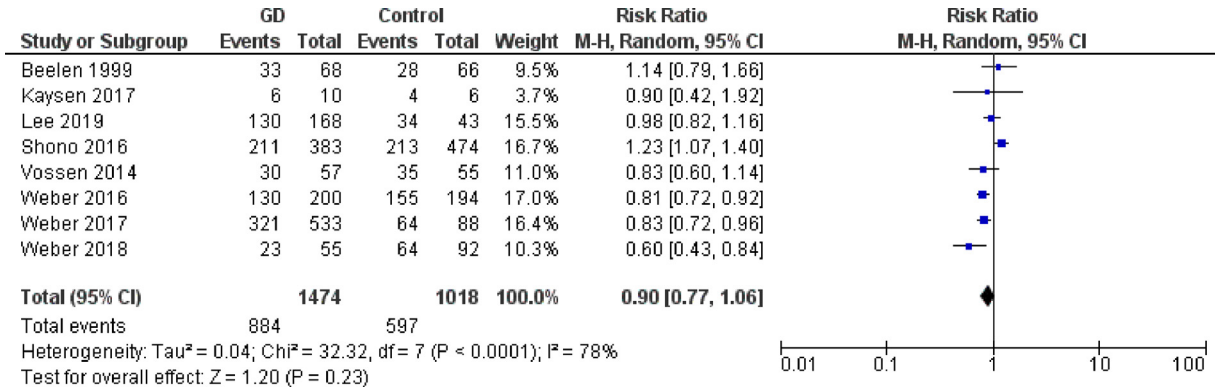


Figure 6. Forest plot of 8 studies with data on overall survival (OS) between patients receiving or not (group versus control) antibiotics. RR and weight of each study are presented along with 95% CI. Analysis was made using the RE model. The diamond represents summary of results and indicates that the overall result is not statistically significant.

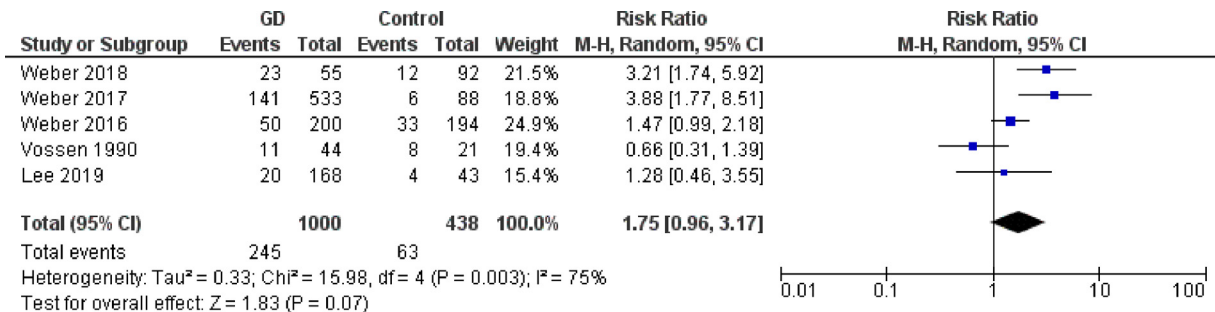


Figure 7. Forest plot of 5 studies with data on TRM between patients receiving or not (group versus control) antibiotics. RR and weight of each study are presented along with 95% CI. Analysis was made using the RE model. The diamond represents summary of results and indicates that TRM is not statistically significantly different between the 2 groups.

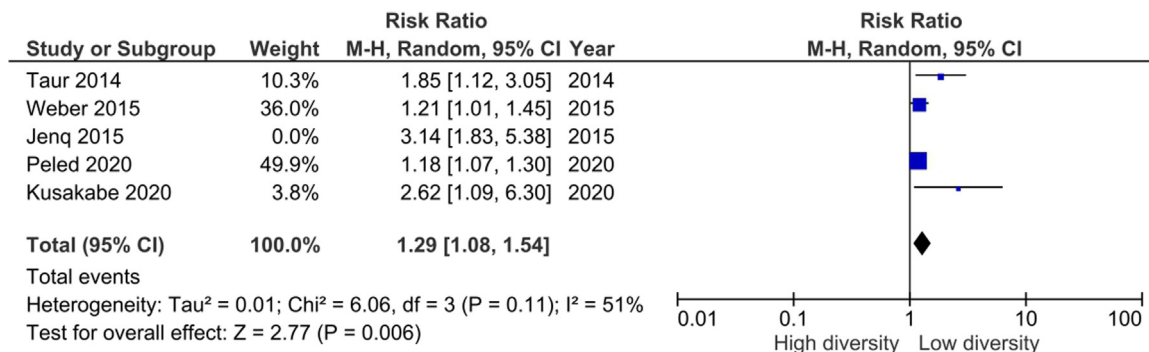


Figure 8. Forest plot of 4 low RoB studies with data on OS between patients with high microbiota diversity versus low. RR and weight of each study are presented along with 95% CI. Analysis was made using the RE model. The diamond represents summary of results and indicates that higher microbiota diversity is associated with better overall survival.

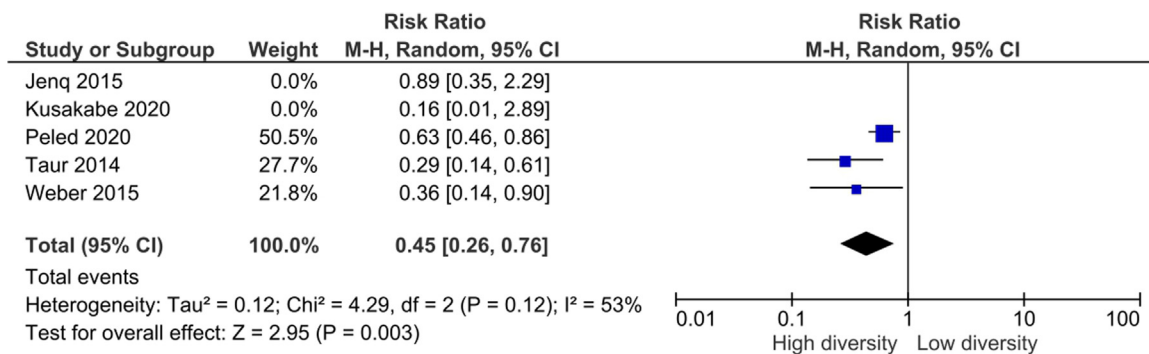


Figure 9. Forest plot of 3 low RoB studies with data on TRM between patients with high microbiota diversity versus low. RR and weight of each study are presented along with 95% CI. Analysis was made using the RE model. The diamond represents summary of results and indicates that higher microbiota diversity is associated with a lower risk of TRM.

Finally, patients with high microbiota diversity had a 55% lower risk of TRM (RR, 0.45; 95% CI, 0.26 to 0.76; $I^2 = 53\%$; $P = .003$; Figure 9). Moderate heterogeneity could be attributed to the abovementioned variety of diversity estimation and to different endpoints (1, 2, and 3 years) of TRM measurement among low RoB studies.

Risk of Bias across Studies and Publication Bias

Quality of evidence was very low applying the GRADE tool (Supplementary Figure S3) [26]. This can be attributed to the observational nature of cohorts included in the meta-analysis, high risk of bias in RCTs, and significant heterogeneity in results that come from observational data and suspected publication bias (asymmetric funnel plot shown in Supplementary Figure S4).

DISCUSSION

Our systematic review and meta-analysis confirm that GD and systemic antibiotics as prophylaxis increase the incidence of acute GVHD based on studies of genomic microbiota diversity. Similarly, GD increases the risk of intestinal GVHD. The power and necessity of the meta-analysis, as a tool enhancing our understanding of the field, is highlighted by the fact that most included studies did not reach statistical significance per se. Importantly, our meta-analysis was the first to show a significant effect of microbiota diversity on TRM and OS.

The only relevant meta-analysis was performed in 2018 by Wang et al. [57]. Although the authors did not take into consideration microbiota markers and did not perform qualitative analysis, this study is in line with our basic findings. Wang et al. [57] showed that patients receiving GD were at higher risk of acute GVHD. Nevertheless, the authors were not able to show a statistically significant difference of OS between the 2 groups. It should be noted that our study evaluated all relevant studies in a systematic way, according to international standards and recommendations, to ensure the reproducibility of our results. The lack of homogeneous and standardized risk scores based on microbiota diversity loss and domination of a single taxon hampers systematic comparisons across different studies. Therefore, we categorized our studies according to outcomes and type of markers. Hence, we proved that disrupted microbiota diversity affected by antibiotics administration is associated with TRM and OS. A predefined microbiota marker or a score, such as the one recently proposed by Han et al. [41] or Peled and Gomes [47], could be used unanimously in future trials as a predictor of allo-HCT outcomes.

Our analysis confirms suggestions of recent studies that have revolutionized the field. An interesting finding, however, is the inverse association observed in the pediatric population,

with GD being associated with less severe acute GVHD (OR, 0.28; 95% CI, 0.1, 0.73; $I^2 = 0\%$; $P = .01$). Pediatric studies that compared GD to controls were performed by Vossen et al. in 1990 [10] and 2014 [33] using cultures. It is well established that this technique does not appreciate the advantages of microbial diversity, as shown by more recent studies. Some of these recent studies could not be included in the meta-analysis due to the absence of a comparative group [45,50,53]. Thus, the above-mentioned result could change by well-designed prospective studies.

Furthermore, we demonstrated associations among antibiotic treatment, altered gut microbiota, and clinical occurrence of intestinal GVHD. We performed a subgroup analysis on systemic versus topical or no antibiotics as prophylaxis. Our results indicate that exposure to systemic antibiotics as prophylaxis should be avoided. A phase II randomized clinical trial is currently under way to evaluate the contribution of broad-spectrum antibiotics and microbiota injury to the development of GVHD by comparing piperacillin-tazobactam with cefepime as a treatment for febrile neutropenia [58]. A randomized phase II clinical trial evaluating the impact of GD with vancomycin plus polymyxin B on microbial diversity after allo-HCT is also ongoing [59]. Despite that our meta-analysis is a powerful tool to support the use of non-broad-spectrum antibiotics, prospective RCTs are needed to indicate the proper prophylactic regimen. Additional strategies to overcome the impact of antibiotics based on probiotics or rapid restoration of microbial diversity early after cessation of antibiotic treatment by either fecal microbiota transplantation or artificially composed microbiota may be a feasible treatment option [60].

Except for GVHD and OS, several other important outcomes after allo-HCT, in particular, bloodstream infections [61,62] and organ toxicity [63], have been associated with intestinal bacterial composition. Despite accumulating evidence of flora's diverse pathophysiologic mechanism on GVHD [64–69], data are limited regarding the composition of favorable microbiota ecologies and allo-HCT [70]. Recent data point toward the protective role of Lachnospiraceae of the class of Clostridia and particularly bacteria belonging to the genus *Blautia* against gastrointestinal GVHD [34,49]. Interestingly, further research led to the observation that high levels of urinary 3-IS were associated with Clostridiales abundance. Hence, the relationship of lower 3-IS levels with worse OS and TRM suggested it as a useful microbiota marker of allo-HCT outcomes [36,39]. In addition, intestinal damage is also observed in other entities within the spectrum of GVHD in allo-HCT recipients. The role of microbiota diversity needs to be further studied in relevant

entities with intestinal damage, such as intestinal thrombotic microangiopathy [71,72].

In addition to strengths described above, our review has also certain limitations. First, the absence of a comparative group did not allow us to include many recent studies in the meta-analysis. In addition, we were not able to compare the effects of certain antibiotics or classes due to the retrospective nature of real-world evidence included in the majority of studies and the lack of a well-designed RCT in the field. This is also depicted at very low certainty of the effect estimate applying the GRADE tool. Furthermore, use of antibiotics as prophylaxis or febrile neutropenia varies across studies. Since this is expected in real-world studies, excluding these studies from our analysis would not confer an accurate reflection of existing data. Last, our analysis was not designed to assess the efficacy of recent strategies under investigation for microbiota diversity outcomes, such as fecal transplantation.

In conclusion, our meta-analysis will be a useful tool for clinical practice and guidance of further research in the field. Further RCTs using standardized markers of microbiota diversity are needed to improve the quality of evidence and certainty in our estimates. Future studies should be performed together with detailed assessment of bacterial susceptibility to antibiotics, as the 16S rRNA sequence cannot provide this information.

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Authorship statement: M.G. and E.G. screened the studies, extracted the data in mutual agreement, and collaborated for the design of the study. M.G. performed the data collection, wrote the first draft of the manuscript, and performed analyses. I.S. and A.A. participated in study conception and design and edited the manuscript.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.bbmt.2020.05.011.

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